

UNIT 3

$$x^T y = y^T x = x_1 y_1 + \dots + x_n y_n \rightarrow \text{inner product.}$$

$$\|x\| = \sqrt{x_1^2 + \dots + x_n^2} \Rightarrow \|x\|^2 = x^T x = x_1^2 + \dots + x_n^2 \rightarrow \text{norm.}$$

~~orthogonal to nullspace do $Ax = 0$ [row space]~~

orthogonal to row space do $Ax = 0$

orthogonal to colspace do $A^T y = 0$.

orthogonal to nullspace is row space

[vectors that span row space]

orthogonal to left nullspace is colspace

[vectors of matrix that span colspace]

orthonormal basis

$\Rightarrow$ normalise the orthogonal vector

$C(A)$ is orthogonal complement of $N(A^T)$

$C(A^T)$ is orthogonal complement of $N(A)$

$$\cos \theta = \frac{a^T b}{\|a\| \|b\|} \rightarrow \text{angle b/w non zero vectors}$$

$$\hat{x} = \frac{a^T b}{a^T a} \quad \bullet \quad \boxed{c = b - p}$$

$$p = \hat{x} a = \frac{a^T b}{a^T a} a \rightarrow \text{projection onto a line.}$$

$\hookrightarrow$ [projection of b onto a]

$$\left| \cos \theta \right| \leq 1 \quad \left. \begin{array}{l} \\ \Rightarrow |a^T b| \leq \|a\| \|b\| \end{array} \right\} \rightarrow \text{Schwarz inequality}$$

$$P = \frac{aa^T}{a^T a} \rightarrow \text{projection matrix used to project } b \text{ onto } a \text{ by } \Rightarrow p = Pb$$

Projection Matrix of Rank 1:

→ symmetric matrix

→ $P^n = P \quad \forall n \in \mathbb{N}$

→ rank / trace of P is one.

→ a - n dimensional vector

P - square matrix of order n .

→ a - unit vector

$$P = aa^T$$

→ P stays the same even if a is doubled.

Transpose:

A^T is reflection of A through main diagonal

$$A_{ij}^T = A_{ji} \rightarrow \text{Transpose by reflection.}$$

$$(Ax)^T y = x^T A^T y = x^T (A^T y)$$

$$(A^{-1})^T = (A^T)^{-1}$$

* Projection P_2 for line $\perp$ to ' a ' can be calculated as $P_2 = I - P_1$ [$P_1 \rightarrow$ projection of a]

$$P_1 + P_2 = I$$

$P_1 \cdot P_2 = 0 \rightarrow P_1$ is orthogonal to P_2

Projections and Least Squares:

- Error $(E) = \|Ax - b\|$

- least squares:

$\rightarrow$ minimise square of dist b/w Ax & b [error]

- Best Approximate Solution:

$$\rightarrow A^T A x = A^T b$$

$\rightarrow$ normal equation.

- Projection vector:

$\rightarrow \hat{p} = \hat{x}a$ where $\hat{x} \rightarrow$ least squares solution.

- Projection Matrix:

$$\rightarrow P = A(A^T A)^{-1} A^T$$

$$\rightarrow p = Pb$$

- Best Estimate:

$$\rightarrow \hat{x} = (A^T A)^{-1} A^T b$$

- * b in column space $p = b$ *
 - * b in left nullspace $p = 0$ *
 - * if A is invertible $p = b$ *
- } vector
- * $P^2 = P$ and $P^T = P$ [matrix, not vector]

Weighted least Squares:

- $A^T W^T W A x = A^T W^T W b \rightarrow$ normal equation
- $C = W^T W$
- Inner Product:
 - $\rightarrow x$ and $y \Rightarrow y^T C x$
- Error:
 - $\rightarrow b - A \hat{x}_w$
- Perpendicularity:
 - $\rightarrow (W y)^T \cdot (W x) = 0$
- Variance:
 - $\rightarrow$ given by the average of squared error
 - $\rightarrow$ weights from variance:
$$w_i = \frac{1}{\sigma_i^2}$$

Orthonormal Basis:

- find the orthogonal basis & divide all vectors by their length to make it unit vector. This makes orthonormal basis.
- vectors $q_1, \dots, q_n$ are orthonormal if:

$$q_i^T q_j = \begin{cases} 0 & \text{when } i \neq j \text{ (orthogonality)} \\ 1 & \text{when } i = j \text{ (normalisation)} \end{cases}$$

- Q: matrix with orthonormal vectors
orthogonal / orthonormal matrix.

Orthonormal Matrices:

$$\rightarrow Q^T Q = I$$

$$\rightarrow Q^T = Q^{-1} \text{ (if square matrix only)}$$

$$\rightarrow Q^T \rightarrow \text{left inverse if } Q \rightarrow \text{rectangular matrix}$$

$\rightarrow$ Permutation matrices are orthogonal.

$\rightarrow$ Multiplying by Q doesn't change length.

$$\|Qx\|^2 = \|x\|^2$$

$$(Qx)^T(Qx) = x^T Q^T Q x = x^T x$$

Projection:

$$\rightarrow p = (q_1^T b) q_1$$

$$\rightarrow Qx = b \Rightarrow x = Q^T b$$

Least Squares:

$$\rightarrow Q^T Q x = Q^T b$$

Gram Schmidt Orthogonalisation:

Step 1: Normalise first vector

$$\rightarrow q_1 = \frac{a}{\|a\|}$$

Step 2: Remove component of b in q_1 direction

$$\rightarrow B = b - (q_1^T b) q_1$$

$$\rightarrow q_2 = \frac{B}{\|B\|}$$

Step 3: Repeat step 2 for c in q_1 & q_2 directions.

$$\rightarrow c = c - (q_1^T c) q_1 - (q_2^T c) q_2$$

$$\rightarrow q_3 = \frac{c}{\|c\|}$$

$$\underline{A = QR}$$

$$A = [a \ b \ c]$$

$$A = [q_1 \ q_2 \ q_3] \begin{bmatrix} q_1^T a & q_1^T b & q_1^T c \\ 0 & q_2^T b & q_2^T c \\ 0 & 0 & q_3^T c \end{bmatrix} = QR$$

Normal Equation :

$$A^T A x = A^T b$$

$$(QR)^T QR x = (QR)^T b$$

$$R^T Q^T QR x = R^T Q^T b$$

$$[Q^T Q = I]$$

$$R^T R x = R^T Q^T b$$

$$\Rightarrow \boxed{R x = Q^T b}$$

- when $Ax = b$ is not solvable, $Rx = Q^T b$ is used.

- If A is square so one Q & R

- If $A = m \times n$ $R = n \times n$
 $Q = m \times n$

Eigenvectors and Eigenvalues :

- Vector that only changes by a scalar factor after applying transformation is eigenvector
- The value attached to the vector is called eigenvalue $[\lambda]$

$$Ax = \lambda x$$

- x is in nullspace of $A - \lambda I$

$$\Rightarrow (A - \lambda I)x = 0$$

$\rightarrow A - \lambda I$ has to be singular

$\rightarrow \det(A) = \lambda_1 \lambda_2 \dots \lambda_n$ (product)

$\rightarrow$ Trace - sum of diagonal elements = sum of eigenvalues

Characteristic Equation :

$$\det(A - \lambda I) = 0$$

Eigenvalues:

- if $A \rightarrow A^k$ then $\lambda_i \Rightarrow \lambda_i^k$
- if $A \rightarrow$ invertible then A^{-1} has eigen values $\frac{1}{\lambda_i}$
- λ_i of $A^T = \lambda_i$ of A
- $A \rightarrow$ diagonal matrix / upper triangular / lower triangular
 $\lambda_i \Rightarrow$ diagonal elements
- If $\lambda_i = 0$ then A is singular [non invertible]

Eigen Vectors:

- If v is an eigenvector for value λ then any non zero scalar multiplied with v is also an eigenvector

$$Av = \lambda v \text{ then,}$$

$$A(cv) = c(Av) = c\lambda v = \lambda(cv) \quad \forall c \neq 0$$

- Any eigenvector has only one eigenvalue.
- Eigenvectors of distinct eigenvalues of matrix are linearly independent
- If matrix is diagonalisable, its eigenvectors form a complete basis for vector space:

$$A = PDP^{-1} \text{ where } P \text{ has eigenvectors.}$$

Caley Hamilton Theorem:

- Every square matrix satisfies its own characteristic equation

Diagonalisation:

→ A is $n \times n$ matrix with ' n ' linearly independent eigenvectors, then

$S \rightarrow$ matrix of eigenvectors as columns
such that:

$$S^{-1}AS = \Lambda$$

where

$\Lambda \rightarrow$ diagonal matrix of eigenvalues.

- Any matrix with distinct eigenvalues can be diagonalised.

$$\rightarrow AS = S\Lambda$$

$$\rightarrow A = S\Lambda S^{-1}$$

$$\rightarrow S^{-1}AS = \Lambda$$

- S is invertible because eigenvectors are linearly independent.

- S is not unique.

- Order of eigenvectors in S is same as order of eigenvalues in Λ .

- Eigenvalues of $A^2 \Rightarrow [\lambda_1^2, \lambda_2^2, \dots, \lambda_n^2]$ and every eigenvector of A is an eigenvector of A^2 .

$$\Lambda^K = S^{-1}A^K S \Rightarrow A^K = S\Lambda^K S^{-1}$$

- If $AB = BA$ & both are diagonalisable then they have the same S .

Eigenspace :

- The eigenspace of a matrix A with eigenvalue λ is set of all eigenvectors associated to λ plus the zero vector.